

Zhuoyuan Li

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Institute for Functional Intelligent Materials
National University of Singapore

Education

2020–2025 **Ph.D.**, School of Mathematical Sciences, Peking University, Beijing, China
co-supervisors: Prof. Pingwen Zhang¹, Prof. Bin Dong²
2016–2020 **B.Sc.**, School of Mathematics, Sichuan University, Chengdu, China

Working Experience

2025–present Research Fellow Institute for Functional Intelligent Materials
National University of Singapore, Singapore

Research Interests

- Deep learning for multiscale modeling
- Numerical weather prediction and data assimilation
- Inverse problems and uncertainty quantification

Publications

🔍 [Google Scholar](#)  [ORCID](#)

† → Equal contribution; * → Corresponding author(s)

Journal Articles

- J1. **Zhuoyuan Li**, Dong*, B. & Zhang*, P. State-observation augmented diffusion model for nonlinear assimilation with unknown dynamics. *Journal of Computational Physics* **539**, 114240. ISSN: 0021-9991. <https://doi.org/10.1016/j.jcp.2025.114240> (2025).
- J2. **Zhuoyuan Li***, Dong, B. & Zhang, P. Latent assimilation with implicit neural representations for unknown dynamics. *Journal of Computational Physics* **506**, 112953. ISSN: 0021-9991. <https://doi.org/10.1016/j.jcp.2024.112953> (2024).

¹homepage: <https://www.math.pku.edu.cn/pzhang/en/>

²homepage: <http://faculty.bicmr.pku.edu.cn/~dongbin/>

Peer-reviewed Conference Proceedings

- C1. Yang[†], P., Feng[†], Y., Chen[†], Z., Wu, Y. & **Zhuoyuan Li**^{*}. *Spend Wisely: Maximizing Post-Training Gains in Iterative Synthetic Data Bootstrapping in Advances in Neural Information Processing Systems* **38** (2025). <https://arxiv.org/abs/2501.18962>.

Preprints

- P1. **Zhuoyuan Li**^{†,*}, Zhao[†], Y. & Li, M. Ensemble Controlled-Flow Filtering for Implicit Data Assimilation. *arXiv preprint arXiv:2607.12975*. <https://arxiv.org/abs/2607.12975> (2026).
- P2. **Zhuoyuan Li**[†], Zhu[†], A. & Li^{*}, Q. Hypothesis-driven construction of mesoscopic dynamics. *arXiv preprint arXiv:2605.16211*. submitted. <https://arxiv.org/abs/2605.16211> (2026).
- P3. Li[†], L., **Zhuoyuan Li**[†], Li[†], S., Zhan, K., Gao^{*}, H., Chen^{*}, C. & Yang^{*}, L. In-context modeling as a retrain-free paradigm for foundation models in computational science. *arXiv preprint arXiv:2604.23098*. submitted. <https://arxiv.org/abs/2604.23098> (2026).
- P4. Wang[†], R., Jiang[†], J., **Zhuoyuan Li**, Liu, T., Wang, Y., Xie^{*}, M. & Piatkevich^{*}, K. D. Fluorescent protein-based ticker tapes for multiplexed recordings of transcriptional histories in single cells in culture and in vivo. *bioRxiv*. accepted. <https://www.biorxiv.org/content/early/2025/09/12/2025.09.08.675004> (2025).
- P5. Huang[†], X., **Zhuoyuan Li**[†], Liu, H., Wang, Z., Zhou, H., Dong^{*}, B. & Hua, B. Learning to simulate partially known spatio-temporal dynamics with trainable difference operators. *arXiv preprint arXiv:2307.14395*. <https://arxiv.org/abs/2307.14395> (2023).

Talks

EASIAM 2026	Aug. 23-27, 2026	TBA (“Minisymposium: MS22 - Learning-Based Scientific Computation: Theory, Algorithms, and Applications”)
AI4X 2026	Jun. 15-19, 2026	“Learning non-equilibrium mesoscopic dynamics with Onsager principle” (Oral)

Teaching

Peking University

Spring 2023	Assistant Instructor, Advanced Algebra (II)
Fall 2022	Assistant Instructor, Advanced Algebra (I)
Spring 2022	Assistant Instructor, Advanced Algebra (II)
Fall 2020	Teaching Assistant ³ , Advanced Mathematics (C)

Please see my homepage for more details.

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³without teaching tasks